Functions and Function Notation

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Function notation**

Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0) = 3$ **b** $f(2) = 1$ **c** $f(-1) = 10$ **d** $f(\frac{1}{2}) = 1$

(a) $= 3$; (b) $= 1$; (c) $= 10$; (d) $= 1$

2 **[Foundational]** • **Target: Function notation**

If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a.

$3a - 7 = 11$, so $3a = 18$ and $a = 6$.

3 **[Foundational]** • **Target: Identifying functions**

Determine whether each equation defines y as a function of x:

a $y = 2x + 1$ Yes — one output for each input.

b $x = y^2$ No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.

c $y = |x|$ Yes — one output for each input.

d $x^2 + y^2 = 9$ No — a circle fails the vertical line test.

(a) Yes — one output for each input.; (b) No — e.g. $x = 4$ gives $y = 2$ or $y = -2$.; (c) Yes — one output for each input.; (d) No — a circle fails the vertical line test.

4 **[Foundational]** • **Target: Identifying functions**

Explain how the vertical line test is used to decide whether a graph represents a function.

If every vertical line crosses the graph at most once, each input x has at most one output y, so the graph represents a function. If any vertical line crosses the graph more than once, it is not a function.

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5 [Foundational] • Target: Combining functions

A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

a Interpret the value $C(0)$ in context. $C(0) = 250$: the registration fee alone, before any tutoring hours.

b Find $C(12)$. $C(12) = 250 + 540 = 790$ riyals.

c Solve $C(n) = 925$ and interpret your answer. $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.

(a) $C(0) = 250$: the registration fee alone, before any tutoring hours.; (b) $C(12) = 250 + 540 = 790$ riyals.; (c) $45n = 675$, so $n = 15$: a total cost of 925 riyals buys 15 hours of tutoring.

6 [Foundational] • Target: Combining functions

Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = \sqrt{x - 4}$, which requires $x - 4 \geq 0$: domain $x \geq 4$.

7 [Foundational] • Target: Piecewise-defined functions

A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:

a $f(-3) = 1$ **b** $f(0) = 4$ **c** $f(1) = 3$ **d** $f(4) = 9$

(a) $= 1$; (b) $= 4$; (c) $= 3$; (d) $= 9$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise-defined functions

A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.

a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB. $C(x) = 60$ for $0 \leq x \leq 5$, and $C(x) = 60 + 15(x - 5)$ for $x > 5$.

b Find $C(3)$ and $C(8)$. $C(3) = 60$ riyals; $C(8) = 60 + 15(3) = 105$ riyals.

c Is C continuous at $x = 5$? Explain. Yes — both pieces give $C(5) = 60$, so there is no jump at $x = 5$.

(a) $C(x) = 60$ for $0 \leq x \leq 5$, and $C(x) = 60 + 15(x - 5)$ for $x > 5$.; (b) $C(3) = 60$ riyals; $C(8) = 60 + 15(3) = 105$ riyals.; (c) Yes — both pieces give $C(5) = 60$, so there is no jump at $x = 5$.

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9 [Intermediate] • Target: Piecewise-defined functions

Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.

$f(x) = -(x - 2) = 2 - x$ for $x < 2$, and $f(x) = x - 2$ for $x \geq 2$. $f(-1) = 3$; $f(5) = 3$.

10 [Intermediate] • Target: Even, odd, and one-to-one functions

Determine whether each function is even, odd, or neither:

a $f(x) = x^4 - 3x^2$ Even — $f(-x) = f(x)$.

b $f(x) = x^3 - x$ Odd — $f(-x) = -f(x)$.

c $f(x) = x^2 + 2x$ Neither — matches neither pattern (e.g. $f(-1) = -1$, $f(1) = 3$, and $-f(1) = -3$).

(a) Even — $f(-x) = f(x)$.; (b) Odd — $f(-x) = -f(x)$.; (c) Neither — matches neither pattern (e.g. $f(-1) = -1$, $f(1) = 3$, and $-f(1) = -3$).

11 [Intermediate] • Target: Even, odd, and one-to-one functions

Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.

a $f(x) = 2x + 5$ One-to-one — every horizontal line meets the graph at most once.

b $f(x) = x^2$ Not one-to-one — e.g. $y = 4$ meets the graph at $x = \pm 2$.

(a) One-to-one — every horizontal line meets the graph at most once.; (b) Not one-to-one — e.g. $y = 4$ meets the graph at $x = \pm 2$.

12 [Intermediate] • Target: Reading function graphs

The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.

From the graph, $f(1) = 1^2 - 2 = -1$. The parabola is increasing for $x > 0$.

13 [Intermediate] • Target: Reading function graphs

Using the graph of $f(x) = x^2 - 2$ above, estimate all x-values where $f(x) = 2$.

Setting $x^2 - 2 = 2$: $x^2 = 4 \Rightarrow x = \pm 2$, matching where the parabola crosses the horizontal line $y = 2$ on the graph.

14 [Intermediate] • Target: Average rate of change from function notation

For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

$\frac{f(3) - f(1)}{3 - 1} = \frac{10 - 2}{2} = 4$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Average rate of change from function notation** • **Cross-Lesson: 5.02 The Dot Product**

A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.

$\frac{h(2) - h(1)}{2 - 1} = \frac{20 - 15}{1} = 5$. Since the average rate of change is positive, the ball's height is increasing on average over this interval.

16 **[Challenging]** • **Target: Difference quotients** • **Cross-Lesson: 5.03 Parametric Equations**

For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.

$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h$.

17 **[Challenging]** • **Target: Difference quotients** • **Cross-Lesson: 6.01 Polar Coordinates and Conversions**

For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.

$\frac{3(x+h) + 2 - (3x + 2)}{h} = \frac{3h}{h} = 3$. This makes sense because a linear function has constant slope 3 everywhere, so its average rate of change over any interval is always exactly that slope.

18 **[Challenging]** • **Target: Inverse functions (introduction)** • **Cross-Lesson: 6.02 Polar Equations and Graphs**

Find the inverse of $f(x) = 3x - 4$, and verify by checking $f(f^{-1}(x)) = x$.

Swap and solve: $x = 3y - 4 \Rightarrow y = \frac{x+4}{3}$. So $f^{-1}(x) = \frac{x+4}{3}$. Check: $f\left(\frac{x+4}{3}\right) = 3\left(\frac{x+4}{3}\right) - 4 = x + 4 - 4 = x$ ✓.

19 **[Challenging]** • **Target: Inverse functions (introduction)** • **Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

Explain why $f(x) = x^2$ (with domain all reals) does not have an inverse function, but $f(x) = x^2$ restricted to $x \geq 0$ does.

$f(x) = x^2$ is not one-to-one over all reals (e.g. $f(2) = f(-2) = 4$), so it fails the horizontal line test and has no inverse function. Restricting to $x \geq 0$ makes it one-to-one, so an inverse ($f^{-1}(x) = \sqrt{x}$) exists.

20 **[Challenging]** • **Target: Identifying functions** • **Cross-Lesson: 7.02 Logarithms and Their Properties**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x + 3)$ cancels for $xeq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function notation

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. **-5 ✓**

Since $x = 3 \geq 3$, use $f(x) = -2x + 1$: $f(3) = -2(3) + 1 = -5$.

22 [Foundational] • Target: Identifying functions

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. **Odd ✓**

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

23 [Foundational] • Target: Combining functions

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \geq 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

c. **4 ✓**

d. 1

Continuity: $2(1) + 3 = 1(1) + k \Rightarrow k = 4$.

24 [Foundational] • Target: Piecewise-defined functions

Classify $f(x) = x^2$ as even, odd, or neither.

a. **Even ✓**

b. Neither

c. Odd

d. Cannot be determined

Applying Functions principles, $f(-x) = x^2$, so f is even. Therefore, the verified correct option is Even.

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25 [Foundational] • Target: Even, odd, and one-to-one functions

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x - 3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x - 3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y - 4}$, so $f^{-1}(x) = 3 + \sqrt{x - 4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x - 4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2 + 4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

26 [Foundational] • Target: Reading function graphs

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down** ✓

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing x with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

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27 [Foundational] • Target: Average rate of change from function notation

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

28 [Foundational] • Target: Difference quotients

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2 ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Inverse functions (introduction)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?

a. -4

b. -2 ✓

c. -1

d. 1

We solve the problem in three steps: 1. **Determine the quadratic function $f(x)$:** The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$:
$f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. **Determine the linear function $g(x)$:** The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

30 [Intermediate] • Target: Function notation

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. **a = 5, b = 3** ✓

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

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31 [Intermediate] • Target: Identifying functions

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?

a. -10

b. -6

c. 6

d. **10** ✓

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for a.
$f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$.
$f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

32 [Intermediate] • Target: Combining functions

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 2

d. Cannot be determined

For a rational function where the degree of the numerator is equal to the degree of the denominator, the limit as $x \to \infty$ is the ratio of the leading coefficients. Therefore, $\lim_{x \to \infty} f(x) = \frac{a}{d}$. We are given that this limit is $\frac{1}{2}$, so $\frac{a}{d} = \frac{1}{2}$. The condition $\lim_{x \to -1} f(x) = \infty$ implies that $x = -1$ is a vertical asymptote. This means that the denominator is zero when $x = -1$, and the numerator is non-zero when $x = -1$. So.

33 [Intermediate] • Target: Piecewise-defined functions

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. **6** ✓

c. Undefined

d. 3

The function is given by $f(x) = \frac{x^2 - 9}{x - 3}$. We can factor the numerator as a difference of squares:
$x^2 - 9 = (x - 3)(x + 3)$. Thus, for $x \neq 3$, we can simplify the function to $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. To find the limit as x approaches 3, we can substitute 3 into the simplified expression:
$\lim_{x \to 3} f(x) = \lim_{x \to 3} (x + 3) = 3 + 3 = 6$. Although the function is undefined at $x = 3$, the limit exists and is 6.
Therefore, as x approaches 3, $f(x)$ approaches 6.

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34 [Intermediate] • Target: Even, odd, and one-to-one functions

Consider the rational function $f(x) = \frac{x^2-4}{x-2}$. What is the value of the function as x approaches 2?

a. **4** ✓

b. 0

c. Undefined

d. 2

The given function is $f(x) = \frac{x^2-4}{x-2}$. We can factor the numerator as a difference of squares: $x^2 - 4 = (x-2)(x+2)$. Therefore, $f(x) = \frac{(x-2)(x+2)}{x-2}$. For $x \neq 2$, we can cancel the $(x-2)$ term, so $f(x) = x + 2$. As x approaches 2, the value of $f(x)$ approaches $2 + 2 = 4$. Although the function is technically undefined at $x = 2$ due to division by zero, the limit as x approaches 2 exists.

35 [Intermediate] • Target: Reading function graphs

For $0 < x < 1$, simplify $\cos(\sin^{-1}(x))$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$ ✓

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1 - x^2$

d. $\sqrt{x^2-1}$

Let $\theta = \sin^{-1}(x)$, so $\sin\theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where $\cos\theta \geq 0$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-x^2}$.

36 [Intermediate] • Target: Average rate of change from function notation

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $(-\frac{\pi}{2}, \frac{\pi}{2})$, range $(-\infty, \infty)$

d. **Domain $(-\infty, \infty)$, range $(-\frac{\pi}{2}, \frac{\pi}{2})$** ✓

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $(-\frac{\pi}{2}, \frac{\pi}{2})$ (restricted principal output).

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37 [Intermediate] • Target: Difference quotients

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$** ✓

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 6.02 Polar Equations and Graphs

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(4)$?

a. **7** ✓

b. 1

c. 5

d. 13

To find the value of $f(4)$, substitute $x = 4$ into the function. $f(4) = \frac{4^2 - 9}{4 - 3} = \frac{16 - 9}{1} = \frac{7}{1} = 7$.

39 [Challenging] • Target: Function notation • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4° ✓

In the right triangle, opposite $= 12$, hypotenuse $= 13$, so the angle is $\sin^{-1}(\frac{12}{13}) \approx 67.4^\circ$.

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40 [Challenging] • Target: Identifying functions • Cross-Lesson: 7.02 Logarithms and Their Properties

A kite string 50 m long is pulled taut, and the kite is 32 m above the hand holding it. Find the angle of elevation of the string, to the nearest tenth of a degree.

a. 50.2°

b. 39.8° ✓

c. 32.6°

d. 44.8°

In the right triangle, opposite $= 32$, hypotenuse $= 50$, so the angle is $\sin^{-1}\left(\frac{16}{25}\right) \approx 39.8^\circ$.

41 [Challenging] • Target: Combining functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1° ✓

In the right triangle, adjacent $= 9$, hypotenuse $= 15$, so the angle is $\cos^{-1}\left(\frac{3}{5}\right) \approx 53.1^\circ$.

42 [Challenging] • Target: Piecewise-defined functions • Cross-Lesson: 7.04 Applications: Growth and Decay

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5° ✓

b. 81.5°

c. 13.5°

d. 17.1°

In the right triangle, opposite $= 3$, adjacent $= 20$, so the angle is $\tan^{-1}\left(\frac{3}{20}\right) \approx 8.5^\circ$.

43 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A vertical pole 6 m tall casts a shadow 9 m long. Find the angle of elevation of the sun, to the nearest tenth of a degree.

a. 33.7° ✓

b. 56.3°

c. 67.4°

d. 38.7°

In the right triangle, opposite $= 6$, adjacent $= 9$, so the angle is $\tan^{-1}\left(\frac{2}{3}\right) \approx 33.7^\circ$.

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44 [Challenging] • Target: Reading function graphs • Cross-Lesson: 8.02 Graphing Rational Functions

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. **35** ✓

Continuity requires $-2x + k$ at $x = 5$ to equal $(5)^2$: $-10 + k = 25$, so $k = 35$.

45 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 8.03 Introduction to Limits

If $f(x) = \sqrt{x - 5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

b. $x \leq -3$ **or** $x \geq 3$ ✓

c. $x \geq 3$

d. All real numbers

Need $x^2 \geq 9$, so $|x| \geq 3$, giving $x \leq -3$ or $x \geq 3$.

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